MODULE 6 · RECOVERY

Lesson 21: Nose-High Upset Recovery

Lesson 21 of 24 · about 8 min

LEARNING OBJECTIVES

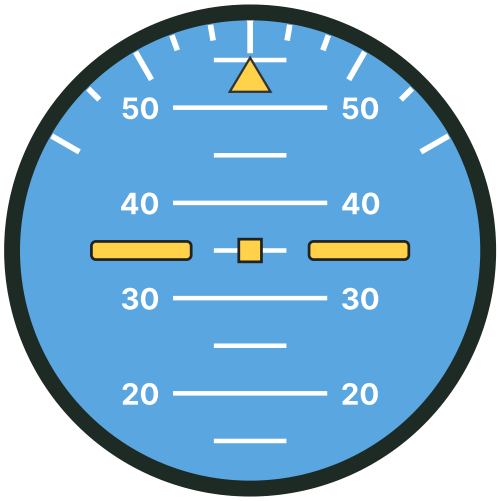
- Apply the generic nose-high recovery technique.
- Explain the use of bank and thrust to lower the nose.
- Complete the recovery without over-correcting into a nose-low upset.



Nose-high: unload, reduce thrust, bank if needed.

The situation

The pitch attitude is unintentionally high — for example more than 25° nose-up — and the airspeed is decreasing, or is already low. The aircraft is heading towards a stall. The goal is to **lower the nose** and regain speed before the angle of attack reaches the critical value.



Nose-high: about 35° up, wings level

Nose-high upset. The speed is decaying; recognise it early.

Generic technique (based on industry guidance)

1. **Recognise and confirm** the situation. Announce “Nose high”.
2. **Autopilot and autothrust — disconnect.**
3. **Apply as much as full nose-down elevator.** Accept reduced g — the feeling of being light in your seat. Use nose-down stabiliser trim if needed.
4. **If more is needed to lower the nose:**
   - **Reduce thrust** (aircraft with underwing engines) to reduce the nose-up moment;
   - **Roll** to a bank angle that helps the nose drop — the lift vector moves away from the vertical, and the nose falls towards the horizon. Do not normally exceed about 60° of bank.
5. **Complete the recovery:** as the nose approaches the horizon, roll wings level, check the airspeed, adjust thrust and establish the required pitch attitude.

Always follow the manufacturer's procedure for your type.

Points to remember

- If the aircraft is **stalled**, apply the stall recovery first (Lesson 20).
- Unloading reduces the angle of attack and the stall speed. A low-g sensation is expected and acceptable.
- Do not over-correct. A nose-high upset recovered too aggressively can become a nose-low, high-speed upset.
- Unsecured objects and passengers or crew without seat belts may be thrown upwards during unloading.

KEY TAKEAWAYS

- Nose-high: automation off, nose-down elevator (as much as full), trim as needed.
- If needed: reduce thrust (underwing engines) and/or roll to bank to help the nose drop — normally not more than about 60°.
- Approaching the horizon: wings level, check speed, set thrust and pitch.

KNOWLEDGE CHECK

In a nose-high upset, full nose-down elevator is not lowering the nose quickly enough. Which additional techniques can help?

A. Add thrust and pull up

B. Reduce thrust and/or roll to a moderate bank angle

C. Extend the flaps

D. Apply alternating rudder

Answer: B. Reducing thrust removes the nose-up moment of underwing engines, and banking lets the nose fall towards the horizon.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.